Adaptive Trust-Aware Retrieval-Augmented Generation for Medical Evidence: Guarding Against Prompt Injection and Retrieval-Induced Hallucination

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***Abstract*—Retrieval-augmented generation (RAG) lets a large language model (LLM) draw on external documents, but retrieval does not make evidence reliable. A medical retriever can return passages that are irrelevant, outdated, attached to the wrong drug, or written to manipulate the model, and a conventional pipeline treats all of them alike once they reach the context window. We built a trust-aware RAG prototype for drug-drug interaction (DDI) and adverse drug event (ADE) questions that holds every retrieved item as untrusted data. Before generation, a chunk is excluded if its provenance check fails, its content hash does not match, it carries instructions addressed to the model, it names none of the requested medications, or its source identity is unverified; the remaining chunks must reach a risk-tiered trust threshold. After generation, each claim is checked by a pinned natural-language-inference (NLI) model at clause level, by a drug-pair identity check, by citation checks and by a medication identity gate, and a claim is released only when all of them agree. We report an offline engineering evaluation. The frozen candidate reproduced byte for byte, with 2,252 backend tests passing in normal and network-blocked modes. On a constructed, author-known set of 113 statements, the identity gate passed none of 54 unsupported statements (95% upper bound 6.6%) and wrongly withheld 2 of 59 valid ones; across seven development sets no unsupported statement was released, and recomputing the release decision from the recorded checks showed that without the identity gate the other checks would have released 3 to 19 identity-unsupported statements per set. 509 scoped security tests passed. Two defects found by tracing withheld statements (a one-way vocabulary rule and a guard that read ordinary words as brand names) were fixed in a remediation candidate, which changed 6 of 731 decisions with no regression. In a small pilot with two hosted LLMs on 30 synthetic medication sets, neither model had an answer released for any of the 10 control sets whose medication pair has no evidence in the corpus, and 1 and 2 answers were released overall; a vision model read 35 of 36 names on the 14 of 20 synthetic prescriptions it answered. The labels are AI-written and unreviewed, the main evaluation used scripted generation, and no comparison with a conventional semantic RAG baseline has been run, so we make no claim about clinical accuracy or about reducing hallucination relative to such a baseline. The prototype is for research and decision support, not for diagnosis or prescribing.**

***Index Terms*—adverse drug events, drug-drug interactions, evidence verification, large language models, prompt injection, retrieval-augmented generation, retrieval poisoning, trust-aware retrieval**

I. Introduction

A. Background and motivation

LLMs are now capable at biomedical question answering and clinical text processing, and their use on medication questions has grown quickly. Their knowledge sits in model parameters, and an answer produced from parameters alone carries no record of where each statement came from. RAG addresses part of this gap: the original formulation paired a parametric generator with a non-parametric document store [1], the approach has grown into a broad family of methods [2], later work added adaptive retrieval and self-critique [3], and medical benchmarks have examined how retrieval choices affect medical question answering [4].

Medicine adds a requirement that general RAG does not address: a passage that is semantically close to the query is not, for that reason, trustworthy. Medication safety shows why. Radha Krishnan et al. compared ChatGPT with pharmacist assessments of DDIs for 120 hospitalised patients and found minimal agreement (Cohen's κ 0.077 to 0.143) [5]. Sicard et al. found that ChatGPT, Claude and Gemini detected most DDI cases in pharmacovigilance reports but had low specificity and reported non-existent interactions for negative controls [6]. Both studies point to the same need: a medication answer is only as safe as the evidence that can be checked behind it.

B. Problem statement

A conventional medical RAG pipeline is

$$Q \;\rightarrow\; \mathcal{R}(Q) \;\rightarrow\; \mathcal{C} \;\rightarrow\; \mathcal{G}(\mathcal{C}, Q), \tag{1}$$

where Q is the query, $\mathcal{R}(Q)$ the retrieved evidence, $\mathcal{C}$ the context assembled from it and $\mathcal{G}$ the generation step. Nothing in (1) tests whether $\mathcal{C}$ is reliable: it may hold irrelevant or outdated evidence, evidence about the wrong medication, low-authority sources, instruction-bearing text or poisoned content. We use the following terms. *Prompt injection* is text intended to manipulate model behaviour; *indirect* injection arrives through retrieved content [7]. *Retrieval poisoning* manipulates the corpus so that misleading content is retrieved

[8]. A *retrieval error* is unsuitable evidence retrieved without an attacker. *Retrieval-induced hallucination* is unsupported or incorrect generated content to which unsuitable retrieved evidence contributed. The research problem is how a medical RAG system can assess the trustworthiness and security of retrieved evidence before that evidence influences the answer, and verify the answer afterwards.

C. Research questions

1) Can evidence-trust assessment make medical retrieval more reliable than semantic relevance alone?
2) Can prompt-injection and retrieval-poisoning controls reduce the influence of malicious retrieved content?
3) Can identity and cross-source checks identify pharmacological claims that are misattributed or weakly supported?
4) Can claim and citation verification reduce unsupported generated statements?
5) Can controlled abstention improve safety when evidence is insufficient or contradictory?
6) What latency do the evidence controls add?

This paper reports evidence on questions 2 to 6 at the level of software behaviour on constructed data (for question 3, its identity side); question 1 and the clinical side of question 3 need data that does not yet exist (Section XI).

D. Contributions

The individual components (trust scoring, adversarial filtering, retrieval, claim verification) are not new on their own. Our contribution is combining them in one implemented pipeline and evaluating it reproducibly:

1) an evidence-eligibility stage that excludes chunks by provenance, content hash, embedded instructions, medication alignment and verified source identity before a nine-factor, risk-tiered trust score is applied;
2) a release rule under which a claim is shown only when its support state, a medication identity gate and a medication guard agree, and no later stage can restore a rejected claim;
3) an evaluation method that reports where each statement stopped (before the gate, at the gate, after it, released) instead of a single accuracy figure, with exact or Wilson intervals, and recomputes the release decision under subsets of the checks from recorded results to isolate what each check adds;
4) a reproducible offline evaluation (frozen datasets, hashes, run records) and a remediation candidate whose two fixes changed 6 of 731 decisions with no regression and no unsupported release;
5) a small pilot of the same release path on output from two hosted LLMs, and of prescription-image extraction with a vision-language model.

II. Related Work

DDI analysis. DDI research has used rules, knowledge graphs and machine learning, and LLMs have recently been evaluated for identifying interactions. ChatGPT's answers depended on prompt wording and agreed only minimally with pharmacists [5]; across three LLMs, detection was high but specificity low, with interactions reported for negative controls [6].

ADEs and pharmacovigilance. Hakim et al. proposed guardrails for LLMs in drug safety: detecting anomalous documents before ingestion, identifying incorrect drug names or adverse-event terms, and conveying uncertainty [9]. Their work supports pipelines in which validated domain information constrains generation.

LLMs for medication safety. Ong et al. evaluated LLM-based decision support built on RAG for prescribing errors across 16 specialties; a pharmacist working with the LLM performed best (accuracy 61%) [10]. Sun et al. grounded LLMs in drug package inserts for medication-safety education; clinical experts scored highest (0.89), ahead of GPT-4o (0.86) [11].

Medical RAG, hallucination and verification. Medical RAG applies RAG [1] to biomedical text [4], where terminology must be normalised [16] and sources differ in authority. Hallucination in generation is well documented [12], and RAGTruth shows that unsupported and contradictory claims persist with retrieval [13]. FActScore checks atomic facts against a source [14]; Self-RAG lets the model critique its own retrieval and output [3]. We instead compute trust and security outside the model. Retrieval uses BM25 [19], dense embeddings [17], [18] and reciprocal rank fusion [20]; verification uses NLI [21] with a biomedical model [22], [23].

Prompt injection and poisoning. Indirect prompt injection compromises real LLM-integrated applications [7], OWASP lists prompt injection among the main LLM application risks [15], and PoisonedRAG showed that a few crafted texts can steer RAG output [8]. Table I summarises what remains open.

TABLE I
Research areas and remaining requirements

Area	Established capability	Remaining requirement
Medical RAG	Retrieval and grounded generation	Evidence trust and adversarial filtering
DDI analysis	Interaction identification	Reliable evidence interpretation
Pharmaco-vigilance	Extraction and guardrails	Integrated retrieval security and provenance
RAG hallucination	Detection and benchmarks	Domain-specific evidence control
Prompt injection	Attacks and defences	Medical-retrieval-aware protection
Retrieval poisoning	Knowledge-corruption attacks	Evidence eligibility and validation
Citation verification	Claim-evidence alignment	Integration with trust and security

III. Threat Model

The adversary can influence the user input, external documents or the retrieval corpus. Table II lists the threats, including non-adversarial failures with similar effects, and the

TABLE II
THREATS AND THE IMPLEMENTED CONTROLS

Threat	Control in the evaluated candidate
Direct prompt injection	Query screening and sanitisation before retrieval
Indirect prompt injection	Chunks carrying instructions for the model are excluded whole
Retrieval poisoning	Provenance poisoning check; poisoning score filter; verified source identity
Evidence tampering	SHA-256 of the text must equal the registered content hash
Entity mismatch	Medication alignment filter; identity gate; drug-pair identity check
Contradictory evidence	NLI stance per clause; aggregate check over retrieved chunks
Citation mismatch	Marker resolution to eligible chunks; alignment and NLI on the cited chunk

implemented control that addresses each. Contradictions occur naturally in the literature and are not treated as attacks by default.

IV. IMPLEMENTED ARCHITECTURE

Fig. 1 shows the components of the frozen candidate (C2h), a Python backend that streams its stages to a browser client. The design treats retrieved information as untrusted until it has passed relevance, trust, security and verification controls. Two simplifications relative to a full production design matter for reading the results: the evaluated corpus is local (no live source routing to PubMed, Europe PMC or openFDA was exercised), and adversarial screening is rule based (no learned detector).

A. Input, identity and retrieval

Medication names are extracted from the request, confirmed by the user and resolved against a pinned RxNorm prescribable lexicon [16] (12,929 entries) and a curated RxCUI cache; brands, ingredients and salts remain distinct concepts. Retrieval runs three channels over the same corpus: BM25 over lower-cased tokens [19], cosine similarity over `all-MiniLM-L6-v2` embeddings [18], and an adjacency-list graph of drug relations seeded with the confirmed medications. Reciprocal rank fusion

$$\mathrm{RRF}(d) = \sum_{r \in \mathcal{L}} \frac{1}{k + \mathrm{rank}_r(d)}, \qquad k = 60, \tag{2}$$

merges the lists [20]; at most 50 fused candidates are kept, chunks with a poisoning score above 0.4 are dropped, and the live path requests the top 20.

B. Evidence eligibility and trust

Each candidate e_i passes five exclusion checks: a provenance poisoning check; equality of the SHA-256 of its text with the registered content hash; an injection screen that excludes the whole chunk (stripping the instruction would change the hashed evidence); medication alignment (the chunk must name a requested medication); and verified source identity. Duplicate texts are kept once. Surviving chunks receive a trust score over nine factors $f_{ik} \in [0, 1]$,

$$T_i = \sum_{k=1}^{9} w_k^{(r)} f_{ik}, \qquad \sum_k w_k^{(r)} = 1, \tag{3}$$

whose weights $w^{(r)}$ depend on the risk tier $r \in \{\mathrm{R0}, \ldots, \mathrm{R3}\}$ of the question (Table III). A factor that the evidence record does not carry is treated as missing and contributes zero, so a gap never raises the score. The eligibility decision is

$$G_i = \big[\,\text{passes all exclusions}\,\big] \wedge \big[\,T_i \geq \tau_r\,\big] \wedge \big[\,A_i \geq 0.3\,\big], \tag{4}$$

with $\tau = (0.30, 0.45, 0.60, 0.75)$ for R0 to R3 and A_i the source authority. Adversarial risk enters as hard exclusions and as the anti-poisoning and anti-injection factors, so a high trust score cannot buy back a security flag. The trust score filters chunks; it does not reorder them, so eligible evidence keeps its fused rank.

TABLE III
TRUST FACTOR WEIGHTS PER RISK TIER (CONFIGURATION OF C2H)

Factor	R0	R1	R2	R3
Source authority	0.15	0.20	0.25	0.30
Query relevance	0.20	0.20	0.15	0.10
Evidence quality	0.10	0.15	0.20	0.25
Freshness	0.10	0.10	0.10	0.10
Consistency	0.15	0.10	0.10	0.10
Entity match	0.10	0.10	0.10	0.10
Population match	0.05	0.05	0.05	0.03
Anti-poisoning	0.10	0.05	0.03	0.01
Anti-injection	0.05	0.05	0.02	0.01
Threshold τ_r	0.30	0.45	0.60	0.75

C. Claim verification and release

Eligible chunks are numbered and passed to the generator as data, and each claim must cite them with `[Source N]` markers. Each generated claim c is then checked four ways. A pinned NLI model (PubMedBERT fine-tuned on MNLI and MedNLI, at a fixed revision [22], [23]) judges each clause against its cited chunk, so one unsupported clause makes the claim unsupported; an aggregate check considers all retrieved chunks. A pair-identity check compares the drug pair, relation and direction with an approved relationship record. Citation checks resolve each marker to an eligible chunk and check alignment and claim strength. The medication identity gate requires every medication or class that the claim puts in a drug role to be named word for word by a cited source (text or verified title), resolving anaphors and rejecting unresolved or invented names; a separate guard rejects medications the user did not confirm. The release predicate is

$$\begin{aligned}\mathrm{Release}(c) = {} & \big[\, s(c) = \textsc{supported} \,\big] \wedge \big[\, g(c) = \textsc{pass} \,\big] \\ & \wedge \big[\, m(c) = \varnothing \,\big],\end{aligned} \tag{5}$$

where s is the final support state, g the gate decision and m the guard message. The predicate reads the gate and guard records

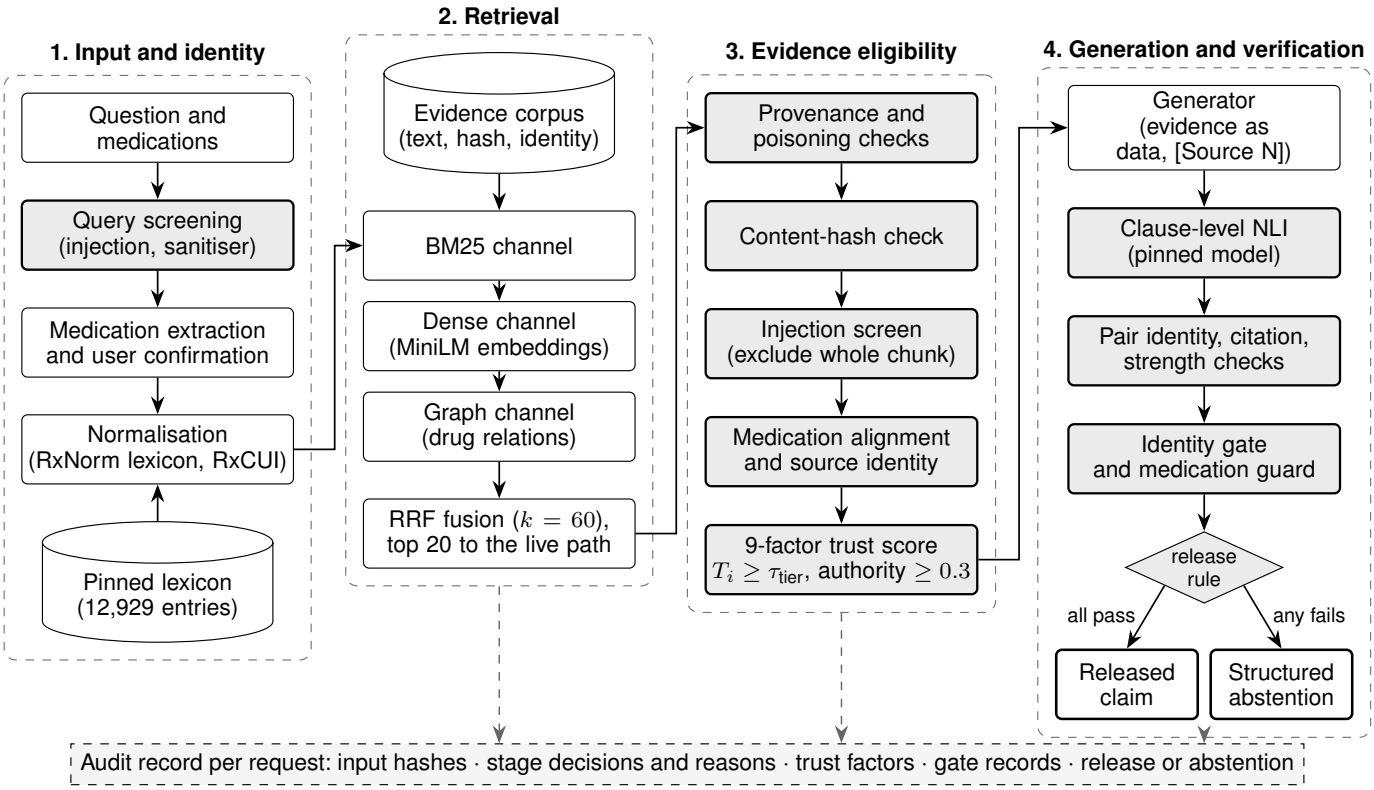


Fig. 1. Implemented architecture of the evaluated candidate (C2h). Shaded nodes are the evidence controls added to a conventional RAG pipeline. Any failed exclusion check removes a chunk; any failed claim check withholds the claim.

directly, so no later stage can restore a rejected claim, and a claim without a gate record fails. Withheld claims produce a structured abstention with the stage and reason.

V. Algorithms

Algorithms 1 and 2 give the two stages as implemented. Admission applies every exclusion before the trust score, so a chunk that fails one is never scored, and an empty evidence pack ends the request with an abstention before any model is called. Verification runs the verifier and the identity gate on every claim, and the strength, citation-alignment and guard checks on every claim the verifier passes, and records each result even when an earlier check has already failed; release requires all of them (5). Because the results are recorded separately, the release decision under any subset of the checks can be recomputed from stored runs (Section IX-D).

VI. Workflow Analysis

The service streams these stages to the client: extracting, normalising, retrieving, security checking, trust evaluating, generating, claim verifying, safety gating and complete. Fig. 2 follows one question. Three properties shape the evaluation. A statement can stop in four places (before the identity gate, at the gate, after it, or released), and counting an upstream abstention as a gate rejection would credit the gate for work it did not do. The checks are conjunctive, so adding one can only reduce what is released: unsupported releases are easy

Algorithm 1 Evidence admission (pre-generation)

Require: query Q, confirmed medications D; tier thresholds τ
Ensure: numbered evidence pack $\mathcal{P}$ or abstention

screen and sanitise Q; resolve D in the pinned lexicon
$\mathcal{E} \leftarrow$ RRF of BM25, dense and graph rankings (2); drop poisoning score > 0.4
$r \leftarrow$ risk tier of Q; $\mathcal{P} \leftarrow \emptyset$
for each $e_i \in \mathcal{E}$ in fused order **do**
  if provenance blocked **or** hash mismatch **or** carries instructions **then continue**
  end if
  if e_i names no medication in D **or** identity unverified **then continue**
  end if
  if text already in $\mathcal{P}$ **then continue**
  end if
  $T_i \leftarrow$ (3) with missing factors set to 0
  if $T_i \geq \tau_r$ **and** $A_i \geq 0.3$ **then** $\mathcal{P} \leftarrow \mathcal{P} \cup \{e_i\}$
  end if
end for
if $\mathcal{P} = \emptyset$ **then return** abstention (no eligible evidence)
end if
return $\mathcal{P}$ numbered as [Source 1..N]

Algorithm 2 Claim verification and release (post-generation)

Require: answer y generated from $\mathcal{P}$; confirmed medications D

Ensure: released claims and abstentions with reasons

1: **for** each claim c in y **do**
2: $s(c) \leftarrow$ clause-level NLI on cited chunks, aggregate check, pair identity, citation and strength checks
3: $g(c) \leftarrow$ identity gate on drug-role names vs. cited text and titles
4: $m(c) \leftarrow$ guard: medications named but not in D
5: **if** (5) holds **then** release c
6: **else** withhold c and record the first failing stage
7: **end if**
8: **end for**

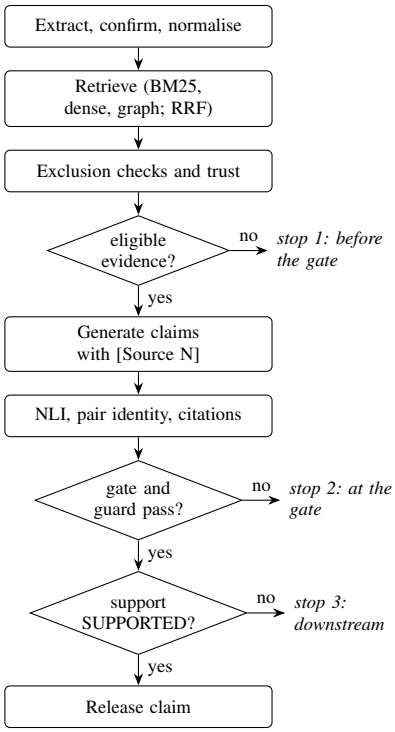


Fig. 2. Workflow of one question and the stop points reported in the evaluation.

to bound and coverage is easy to lose. Every stage writes a record, so each withheld statement can be traced to the first stage that stopped it.

VII. Data and Evidence Management

Every chunk carries a provenance record: source type, document identifier, URL, retrieval time, content hash and, where established, a source-identity record with title and version. The evaluated local corpus has 17 items. All 17 content hashes match their text, but only the three DailyMed label sections (lisinopril sections 5.5 and 7.1, warfarin sodium section 7.3) have a verified source identity; the other 14 chunks have none and 11 carry placeholder URLs, and no relationship record has owner approval. A matching hash shows that stored text is unchanged; it does not show that the text came from the claimed source. DailyMed shows each product's in-use labeling, which may differ from the latest FDA-approved labeling, so set identifier, version, section and retrieval date are kept with each item.

TABLE IV
Development sets (frozen; SHA-256 in the dataset registry)

Set	Scored	PASS	REJECT	Role
V2	119	79	40	regression
V3	101	65	36	regression
V4	72	44	28	regression
V5	84	48	36	regression (88 rows)
V6	126	65	61	regression
V7	116	55	61	regression
V8	113	59	54	author-known

VIII. Evaluation Method

Apart from the pilot in Section IX-H, all runs used an isolated copy of the frozen candidate, offline, with the pinned NLI model and no provider call. Generation was scripted: each test statement is fed through the real release path as if a model had written it, so the checks are measured, not a language model. Seven constructed statement sets (V2 to V8; Table IV) were written during development, each after the previous candidate had been tested, so they are development-informed regression sets; V8 was written by the author of C2h and is author-known. Each row is labelled PASS (every medication or class in a drug role is named by the cited source) or REJECT, with the expected failing entity. All labels are AI-written and unreviewed; blind review packets with opaque identifiers are prepared.

For each set we report the gate miss rate (REJECT rows the gate passed, over REJECT rows that reached it), unsupported releases (REJECT rows released), the identity false-withhold rate (PASS rows the gate rejected, over PASS rows) and coverage (PASS rows released). Rates carry 95% intervals: Clopper and Pearson's exact interval [25] when the count is 0 and Wilson's interval [24] otherwise. Paired comparisons on the same rows use the exact McNemar test [26]. Sets are never pooled into one rate. The acceptance criterion for identity false withholds, fixed before any new confirmation set, is a point estimate of at most 5% with the Wilson interval reported.

IX. Results

A. Reproducibility

A clean offline rerun of the frozen candidate produced 15 of 15 result files byte-identical to the stored reference, and the backend suite gave 2,252 passed, 0 failed and 4 skipped (provider credentials absent) in normal and network-blocked modes (run `C2H_REPRO_01`).

B. Identity gate

Table V and Fig. 3 give the per-set results. No REJECT row was released in any set. Among REJECT rows that reached the gate, the gate passed one (V6-IND-03, a row whose label is a

TABLE V
IDENTITY GATE PER DEVELOPMENT SET (C2H; RUNS STOP_POINTS_SUPPLEMENTARY_01, OFFLINE_METRICS_01)

Set	Gate miss (95% CI, %)	Not reached	False withhold (95% CI, %)	Coverage
V2	0/40 (0, 8.8)	0	0/79 (0, 4.6)	45/79
V3	0/36 (0, 9.7)	0	2/65 (0.9, 10.5)	40/65
V4	0/28 (0, 12.3)	0	0/44 (0, 8.0)	31/44
V5	0/36 (0, 9.7)	0	0/48 (0, 7.4)	26/48
V6	1/61 (0.3, 8.7)	0	1/65 (0.3, 8.2)	34/65
V7	0/60 (0, 6.0)	1	0/55 (0, 6.5)	13/55
V8	0/54 (0, 6.6)	0	2/59 (0.9, 11.5)	17/59

Unsupported releases: 0 in every set. Labels AI-written; sets development-informed.

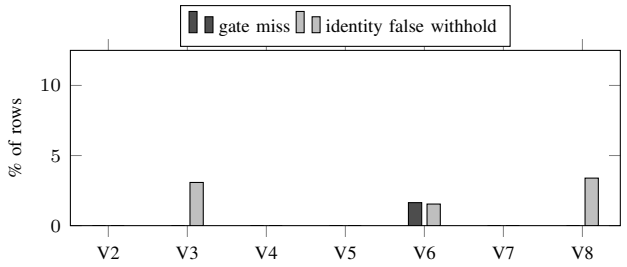


Fig. 3. Gate misses and identity false withholds per development set (C2h). Interval widths are in Table V.

TABLE VI
FIRST WITHHOLDING STAGE FOR VALID STATEMENTS THAT PASSED THE GATE (V2 TO V8; RUN STOP_DIAGNOSTICS_C2H_01)

Stage	Rows	Recorded detail
NLI: cited chunk does not entail	129	entailment < 0.10 in 101
Pair identity unavailable	52	34 claim, 18 source unresolved
Verifier aggregate	18	insufficient over all chunks
Medication guard	5	ordinary words flagged

recorded error). One REJECT row (V7-SRC-02) never reached the gate: its cited section names neither requested medication, so the pipeline abstained before generation. On V8 the two false withholds came from the ordinary words "nudge" and "oozing", which the gate's word list did not contain.

C. Where valid statements stop

Coverage is low (Fig. 4). Across V2 to V8, 206 valid statements were released, 5 were rejected by the identity gate and 204 passed the gate and were withheld later (Table VI). The pair-identity stops follow from missing or unapproved relationship records (34 claims whose canonical identity was not resolved, 18 whose source identity was unavailable), so they are correct under the current evidence. Whether the NLI and aggregate stops are right cannot be decided without claim-support labels. The five guard stops are a defect: the guard reported the ordinary words "Perform" (three times), "Align" and "sodium" as unconfirmed medications although the gate had passed them. "Perform" and "Align" are RxNorm brand names opening an instruction ("Perform potassium monitoring periodically"), and "sodium" occurs in "a low-sodium diet".

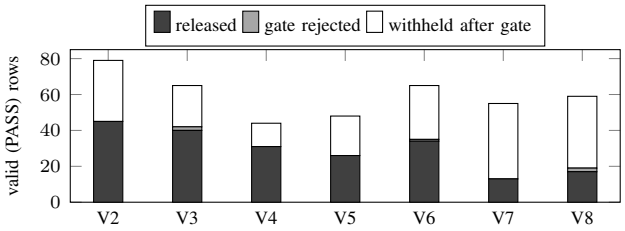


Fig. 4. Stop points of valid statements per development set (C2h; run OFFLINE_METRICS_01).

TABLE VII
ROWS RELEASED UNDER SUBSETS OF THE CHECKS, RECOMPUTED FROM RECORDED RESULTS (C2H; RUN CHECK_COUNTERFACTUAL_01)

	REJECT rows released				PASS rows released		
Set	none	gate only	all but gate	all	of	gate only	all
V2	40/40	0	14	0	79	79	45
V3	36/36	0	14	0	65	63	40
V4	28/28	0	5	0	44	44	31
V5	36/36	0	3	0	48	48	26
V6	61/61	1	19	0	65	64	34
V7	61/61	0	10	0	55	55	13
V8	54/54	0	10	0	59	57	17

"None": no eligibility stage and no check, as in (1); the other columns keep the eligibility stage, which stopped V7-SRC-02. Scripted generation; labels AI-written; sets development-informed.

D. What each check contributes

Because every check result is recorded (Section V), we recomputed the release decision of every development row under subsets of the checks (Table VII; run CHECK_COUNTERFACTUAL_01). Recomputing the full rule reproduced the recorded outcome on all 731 rows. A conventional pipeline (1) fed the same scripted statements shows all of them, including every REJECT row. With every check except the identity gate, 3 to 19 identity-unsupported statements per set would have been released (75 rows in the seven sets). In these rows the NLI model found the claim entailed by its cited chunk, but the chunk does not name a medication or class that the claim asserts: invented names ("Duskwell can raise the bleeding hazard for warfarin users") and classes that the warfarin or lisinopril section does not name ("SSRIs may call for closer INR follow-up in warfarin users"). The identity gate alone released one REJECT row (V6-IND-03, the recorded label error) and withheld no valid row apart from its own false withholds, but it checks no semantic support; the full rule released 13 to 45 valid rows per set. The sets were written to exercise the gate, so these counts show what the other checks miss on such statements, not how often such statements occur in real answers.

E. Candidate comparison and remediation

Fig. 5 compares three code versions on the same 113 V8 statements. The C2h code fixed 10 decisions of the previous candidate (C2g) and broke none (exact McNemar $p = 0.002$); since V8 is author-known, this is exploratory.

TABLE VIII
DECISIONS CHANGED BY THE C2i FIXES (ALL 731 DEVELOPMENT ROWS; RUN C2H_VS_C2I_02)

Row	Label	C2h	C2i	Fix
H-CWD-04	PASS	withheld (guard)	released	guard
H-CWD-08	PASS	withheld (guard)	released	guard
V3-CWD-03	PASS	withheld (guard)	released	guard
V5-CW-04	PASS	withheld (guard)	released	guard
V6-NPH-04	PASS	withheld (guard)	released	guard
V8-SRC-08	PASS	gate reject	withheld (pair id.)	vocabulary

Other 725 rows unchanged; unsupported releases 0 in both.

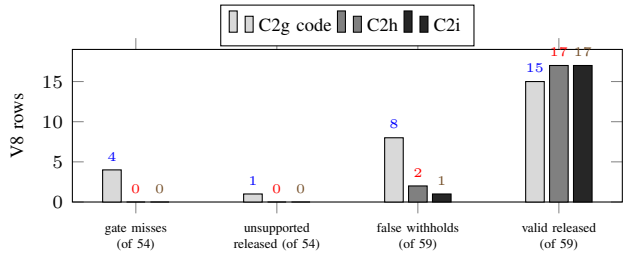


Fig. 5. Three code versions on the same V8 statements (runs V8_PAIRED_C2G_C2H_01, V8_RISK_COVERAGE_01, C2H_VS_C2I_02). V8 is author-known; these are development comparisons, not a baseline comparison.

We then built a remediation candidate (C2i) from C2h to fix the two defects found above without loosening any check. First, an ordinary verb is now known when its listed past or present participle is (the word list held "nudged" but not "nudge"); plural forms are not used, so a name whose plural happens to be listed is not made known. Second, the guard skips an unconfirmed finding only in two forms: an RxNorm brand name that is an ordinary English word, opens a sentence and is not followed by a modal or auxiliary ("Perform potassium monitoring", but not "Perform may raise …"), and an ingredient inside a "low-", "high-" or "reduced-" diet compound. A brand used anywhere else, a brand that is not an ordinary word ("Advil"), an everyday-word ingredient such as lithium, and every confirmed or unresolved name are still reported; 52 new unit tests pin these cases. Table VIII lists every decision that changed. Across all 731 development rows six decisions changed, none for the worse: the five guard rows are now released, and the "nudge" statement passes the gate but is still withheld by pair identity. No unsupported statement was released, the 42-example NLI replay was unchanged, the 509 security tests passed and the backend suite gave 2,304 passed, 0 failed and 4 skipped with the network blocked (run C2I_REGRESSION_02). "Oozing" remains unfixed because no vocabulary we could audit offline contains any form of "ooze". V2 to V8 informed these changes, so these are development-set results.

F. Security regression

We grouped 509 existing tests by threat and ran them with the network blocked (Fig. 6). All passed on C2h, on C2i and on C2j. The tests check specified behaviours with inputs written by the developers; they are not an attack-success rate, and a 110-case security set stays unrun until people review its labels.

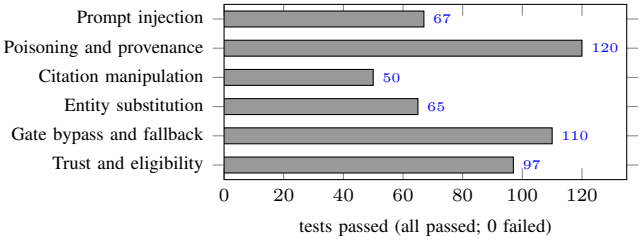


Fig. 6. Scoped security regression by threat group (runs SECURITY_REGRESSION_01, C2J_PORT_CANCELLATION_01).

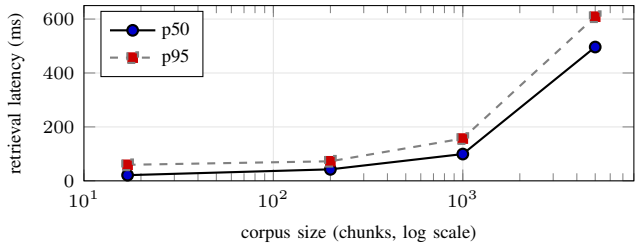


Fig. 7. Hybrid retrieval latency on synthetic corpora of growing size (C2h retriever, one CPU machine, warm process; run PF01_SCALING_01).

G. Performance

On one CPU machine with a warm file cache and the 17-item corpus, median latency was 7.9 ms for retrieval ($n = 63$), 1.4 ms for the identity gate per statement ($n = 635$) and 164.6 ms for the full offline release path ($n = 339$; 95th percentile 336.7 ms); peak working set was 837 MB (run PF01_OFFLINE_LATENCY_02). These figures exclude any language-model call. On C2j (PF01_C2J_SUPPLEMENT_01) one NLI pair took 254.9 ms (median; p95 478.9 ms), local OCR 1.05 s per image, grounding under 2 ms and name normalisation 0.03 ms.

To see how retrieval scales, we built the unmodified hybrid retriever over synthetic, fictional corpora of 17, 200, 1,000 and 5,000 chunks and timed the same 30 queries after 5 warm-up queries (Fig. 7; run PF01_SCALING_01). Median latency grew from 21.0 ms to 42.6, 99.3 and 496.2 ms, roughly linearly with corpus size, because the dense channel scores every chunk by brute-force cosine similarity; building the index (tokenising and embedding) took 1.2 s, 7.7 s, 39.1 s and 183.7 s, and peak working set reached 980 MB at 5,000 chunks. This harness encodes each query with the sentence-transformer directly, so its 17-chunk figure is not comparable with the 7.9 ms above. A production corpus would need an approximate nearest-neighbour index; the synthetic text says nothing about retrieval quality.

H. Pilot with real language models

Everything above used scripted generation. To see how the release path handles text a model actually writes, we ran

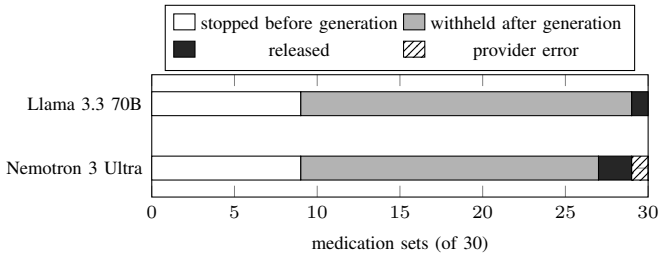


Fig. 8. Terminal outcome per medication set in the real-model pilot (runs LIVE_TEXT_01, LIVE_TEXT_03).

a small authorised pilot on the remediation candidate (C2i) with two hosted models: Llama 3.3 70B Instruct (FP8) on Cloudflare Workers AI and Nemotron 3 Ultra (550B total, 55B active parameters) on NVIDIA's API. Each received the same 30 synthetic medication sets once: 15 with warfarin, 5 with lisinopril and 10 control sets whose medication pair has no evidence in the corpus. The evidence was the three verified DailyMed label sections; only synthetic inputs and those public sections were sent, on free tiers, and the NLI model ran locally. Our first NVIDIA choice, the application's default Nemotron 3 Super 120B, and a second model, Llama 3.1 Nemotron 70B, were both reported as not available to the project's account (21 errors each; runs LIVE_TEXT_01, LIVE_TEXT_02), so Nemotron 3 Ultra replaced them.

Table IX and Fig. 8 give the outcomes. In each run nine control sets stopped before generation because no eligible evidence named their medications, so 21 cases per model reached the language model. The tenth control (sertraline with omeprazole) reached both models because the warfarin section names sertraline; both answers were withheld, and each contained the unsupported claim that the two drugs do not interact. No control set had an answer released. The release rule is all or nothing: a single claim that fails withholds the whole answer. Cloudflare's model had one answer released (10 claims, for lisinopril with metformin) and Nemotron 3 Ultra two (one claim each, for warfarin with ibuprofen requested by the brand names Coumadin and Advil, and for warfarin with atorvastatin). Most failed claims could not be matched to an approved drug-relationship record, the same stop that dominated the scripted runs; Llama 3.3 also produced 39 claims that the identity gate rejected, against 14 for Nemotron. In two Llama 3.3 cases and three Nemotron cases a single failing claim withheld an answer whose other claims all passed. We read the 12 released claims ourselves: each restates its cited DailyMed section (hyperkalaemia, and an excessive fall in blood pressure when lisinopril is started in patients on diuretics; bleeding risk when warfarin is combined with interacting drugs such as NSAIDs), and the lisinopril answer says nothing about metformin, for which the corpus has no evidence. This check was not independent.

The application reads prescription images with a vision-language model and asks the user to confirm each extracted name before retrieval. We drew 20 synthetic prescriptions, five

TABLE IX
REAL-MODEL PILOT ON 30 SYNTHETIC MEDICATION SETS, ONE RUN PER MODEL (C2I; RUNS LIVE_TEXT_01, LIVE_TEXT_03, SCORED IN LIVE_SUMMARY_01)

	Llama 3.3 70B (Cloudflare)	Nemotron 3 Ultra (NVIDIA)
Stopped before generation	9	9
Withheld after generation	20	18
Answer released	1	2
Provider error	0	1 (timeout)
Control sets released	0/10	0/10
Claims generated	141	171
Claims passing every per-claim check	51	82
Claims released	10	2
Failed claims without a relationship record*	38	53
Failed claims with a different relationship*	4	9
Failed claims rejected by the identity gate*	39	14
Time per generated case, median / p95 (s)	9.9 / 30.6	20.2 / 87.8

*As recorded in the abstention reason; one claim can carry several reasons.

each printed plainly, with brand and generic names, rotated, and with added noise, and sent each once (Table X). The application's default vision model, Llama 3.2 11B Vision, had been retired by the provider and answered all 60 attempts with HTTP 410 (LIVE_VISION_01); Llama 3.2 90B Vision timed out on its first three images and we stopped that run (LIVE_VISION_02). Nemotron 3.5 Super VL (120B total, 12B active) answered 14 of 20 images (LIVE_VISION_04). On those 14 it read 35 of the 36 medication names and returned no unrelated name; 13 images were read exactly. The one error matters for this system: on a rotated image it returned "metoprolol" for metoprolol tartrate, dropping the salt, which the confirmation step would have to catch. The other six replies were empty or not valid JSON and failed closed, with no medication passed on; three of them were rotated images. Median time per image was 7.1 s and the 95th percentile 32.7 s with a 60 s timeout. On this evidence we made Nemotron 3.5 Super VL the default vision model in C2i; an explicit setting still overrides it. After this change the full C2i regression gave the same V2 to V8 decisions, an unchanged 42-example replay, 509 of 509 security tests and 2,307 backend tests passed, 0 failed and 4 skipped with the network blocked (three new tests; run C2I_REGRESSION_03); the same count holds in normal mode (run C2J_BASELINE_REPRO_02).

These pilots are small (one run per model, 30 text cases, 20 images), the cases and images were written by us, and the released claims were checked only by us. On real model output the release path behaved much as it did on scripted statements: it withheld most answers and released none for the controls. The pilots do not measure answer quality, and the provider availability problems we hit are themselves a deployment finding.

I. Hardening of the remediation candidate (C2j)

The final candidate, C2j, adds four changes to C2i, ported from a parallel line of the same application (Table XI); none

TABLE X
PRESCRIPTION-IMAGE EXTRACTION, 20 SYNTHETIC IMAGES, ONE ATTEMPT EACH (NVIDIA API; RUN LIVE_VISION_04 UNLESS STATED)

Model and image type	Images	Answered	Exact	Failed closed
Llama 3.2 11B Vision (_01)	20	0	–	20 (HTTP 410)
Llama 3.2 90B Vision (_02)	3	0	–	3 (timeout)
Nemotron 3.5 Super VL				
plain	5	3	3	2
brand and generic	5	4	4	1
rotated	5	2	1	3
noisy	5	5	5	0
all	20	14	13	6

Names on answered images: 35 of 36 read; 1 returned without its salt.

TABLE XI
C2J CHANGES, EACH FOLLOWED BY A FULL REGRESSION: 0 FAILED, 4 SKIPPED IN BOTH MODES; 509/509 SECURITY TESTS (BASELINE C2I: 2,307 PASSED, C2J_BASELINE_REPRO_02)

Change	Run C2J_PORT_	Passed
Single-worker startup guard	SINGLE_WORKER_02	2,316
Provider error bodies not logged	ERROR_BODIES_01	2,322
OCR-first reading, name grounding	OCR_FIRST_02	2,377
Cancellation and request lifecycle	CANCELLATION_01	2,415

touches retrieval, trust scoring or verification. The server now refuses to start with more than one worker, and provider error bodies, which could carry prescription text or the account identifier, are no longer logged. Each uploaded image is read by local OCR first and goes to the hosted model only if that OCR finds text, and a returned name is offered for confirmation only if the OCR text contains it. A cancel endpoint, a disconnect handler and a 120 s deadline for unopened streams end an abandoned analysis and stop its OCR process. After each change the full regression kept all 731 V2 to V8 decisions and the 42-example replay byte-identical to C2i, so the development results above hold for C2j; a comparison of function bodies found the release rule and every verification call unchanged. The frontend passed 148 of 148 tests with a clean build (C2J_FRONTEND_02), and 154 tests run with the real NLI model loaded passed (C2J_REAL_NLI_SUITE_02).

An offline replay of the vision pilot's recorded answers (C2J_FINAL_METRICS_01) found OCR text on all 20 images (median 143 characters), so all would still be sent; grounding kept all 35 correct names and also "metoprolol" for metoprolol tartrate, whose missing salt only the user's confirmation can catch.

X. IMPLEMENTATION STATUS

Table XII records the state of each component of the evaluated candidate. "Tested" means covered by the passing backend suite; "evaluated" means measured on the development sets; "validated" would require independent data and reviewers, and no component has reached it.

TABLE XII
COMPONENT STATUS OF THE EVALUATED CANDIDATE (C2H; C2I AND C2J WHERE MARKED)

Component	Impl.	Tested	Eval.	Valid.
Query screening	yes	yes	no	no
Medication normalisation	yes	yes	partly	no
Hybrid retrieval and RRF	yes	yes	no	no
Exclusion checks (hash, injection, id.)	yes	yes	no	no
Trust scoring and tier thresholds	yes	yes	no	no
Clause-level NLI verification	yes	yes	partly	no
Pair identity and citation checks	yes	yes	partly	no
Medication identity gate	yes	yes	yes	no
Release rule and abstention	yes	yes	yes	no
C2i vocabulary and guard fixes	yes	yes	yes	no
Prescription image extraction	yes	yes	pilot	no
C2j: OCR-first reading, name grounding	yes	yes	replay	no
C2j: cancel, worker guard, log redaction	yes	yes	no	no
Evidence-source adapters (not live)	yes	yes	no	no
Learned adversarial detector	no	no	no	no

XI. LIMITATIONS AND FUTURE WORK

The results describe software behaviour on constructed data. No qualified person has reviewed any label, and V2 to V8 were written during development, so the identity results are preliminary and do not show generalisation; an independently written, blind-reviewed confirmation set is needed. Generation was scripted except in the pilot, which used one run per model on 30 cases and 20 images that we wrote ourselves, so real model output is barely sampled. The local corpus is not a gold corpus (14 of 17 chunks without verified identity), so retrieval quality, claim-level support and citation support remain unmeasured, and so does the comparison with semantic-similarity RAG that research question 1 asks for. The security tests do not measure robustness to unseen attacks, and latency was measured on one machine. The C2j changes were checked by tests and an offline replay only, not with users or real prescriptions. Table XIII summarises what the evidence supports.

The next evaluation is designed but not run. A frozen corpus of at least 200 verified chunks from at least 30 documents, with source identity, version, retrieval date and hash, will be built once source access and reuse terms are approved. Four retrieval configurations will be compared on the same corpus and queries: dense only, BM25 only, dense plus BM25, and hybrid plus the trust controls (with and without the graph channel). Two independent annotators will grade relevance (0, 1, 2) blind to system and rank, with adjudication and weighted κ; we will report P@5, P@10, R@5, R@10, MRR and nDCG@5/10 with query-level bootstrap intervals and paired differences. The metric code already passes 15 hand-calculated synthetic tests. Real-model evaluation and clinical review will follow only with the corresponding authorisation and qualified reviewers.

TABLE XIII
STATUS OF THE PAPER'S CLAIMS

Claim	Status
Offline reproducibility of the frozen candidate	supported
Scoped security regression (509 tests)	supported
Provenance completeness of the corpus (descriptive)	supported
Stop-point diagnosis of withheld statements	supported
Contribution of each check (recomputed)	preliminary
Identity gate results on development sets	preliminary
C2i fixes on development sets	preliminary
Coverage and abstention behaviour	partial
Offline latency (one machine)	supported
Retrieval quality; claim and citation support	blocked
Independent generalisation	blocked
Fewer hallucinations than semantic RAG	not supported
Clinical correctness or safety	not supported
Release path on real model output	pilot only
Prescription image extraction	pilot only
Engineering hardening of C2j (tests, run records)	supported

XII. CONCLUSION

The prototype separates medication identity, provenance and semantic support, combines them in a release rule that no later stage can override, and records where each statement stops. On constructed development data its identity gate released no unsupported statement, at the price of low coverage that mostly comes from checks after the gate, and recomputing the release decision from the recorded checks showed that the gate catches misattributed statements that semantic entailment alone accepts. Whether trust-aware retrieval beats semantic retrieval is still open and needs the verified corpus, independent confirmation set and human relevance judgments described above. The system is a research prototype and is not intended for clinical use.

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